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**IE.3507 – SAFETY AND RELIABILITY**

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# **MTBF Analysis**



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SAKTHIVEL BALAKUMAR - 63957

SELVAN RAGHUNANDAN - 63900

CHIPIRIPALLI SAI DEVIKA - 63923

## TABLE OF CONTENTS

|                                                                                                        |           |
|--------------------------------------------------------------------------------------------------------|-----------|
| <b>INTRODUCTION.....</b>                                                                               | <b>2</b>  |
| <b>CHEMICAL CAPACITOR.....</b>                                                                         | <b>2</b>  |
| <b>RESULTS AT DIFFERENT TEMPERATURES.....</b>                                                          | <b>3</b>  |
| <b>REMARKS</b>                                                                                         |           |
| <b>A) Environments at a fixed temperature (<math>T = 85^{\circ}\text{C}</math>) .....</b>              | <b>6</b>  |
| <b>B) Temperatures in a fixed environment (<math>\text{Env} = \text{GB}</math>) .....</b>              | <b>6</b>  |
| <b>REMARKS.....</b>                                                                                    | <b>6</b>  |
| <b>VOLTAGE ANALYSIS – VOLTAGE IMPACT ON CHEMICAL CAPACITOR (CSR, TANTALUM, SOLID ELECTROLYTE).....</b> | <b>7</b>  |
| <b>REMARKS.....</b>                                                                                    | <b>7</b>  |
| <b>CERAMIC CAPACITOR.....</b>                                                                          | <b>8</b>  |
| <b>RESULTS AT DIFFERENT TEMPERATURES.....</b>                                                          | <b>8</b>  |
| <b>REMARKS</b>                                                                                         |           |
| <b>A) Environments at a fixed temperature (<math>T = 45^{\circ}\text{C}</math>).....</b>               | <b>11</b> |
| <b>B) Temperatures at a fixed environment (<math>\text{Env} = \text{NS}</math>).....</b>               | <b>11</b> |
| <b>REMARKS.....</b>                                                                                    | <b>12</b> |
| <b>TRANSISTOR RELIABILITY ANALYSIS.....</b>                                                            | <b>12</b> |
| <b>Objective.....</b>                                                                                  | <b>12</b> |
| <b>Methodologie.....</b>                                                                               | <b>12</b> |
| <b>Parameters.....</b>                                                                                 | <b>13</b> |
| <b>Results and Discussion.....</b>                                                                     | <b>13</b> |
| <b>A) Temperature Factor Calculation (<math>\pi_T</math>).....</b>                                     | <b>13</b> |
| <b>B) Reliability Results for All Environments:.....</b>                                               | <b>14</b> |
| <b>REMARKS.....</b>                                                                                    | <b>15</b> |
| <b>Temperature Effect.....</b>                                                                         | <b>15</b> |
| <b>Environment Effect.....</b>                                                                         | <b>15</b> |
| <b>SUMMARY.....</b>                                                                                    | <b>15</b> |
| <b>CONCLUSION.....</b>                                                                                 | <b>16</b> |

## INTRODUCTION:

In electronic system design, **reliability analysis** plays a vital role in predicting how long components can operate before failure. Every electronic component is affected by factors such as **temperature, environment, and electrical stress**, which gradually reduce its performance and lifetime.

To estimate this lifetime, the **Mean Time Between Failures (MTBF)** method is used. It provides a statistical measure of the average operating time between two consecutive failures, helping engineers evaluate and improve product dependability. The **MIL-HDBK-217F reliability prediction tool** is widely used in the electronics industry to calculate the **failure rate ( $\lambda_p$ )**, **Failures in Time (FIT)**, and **MTBF** of components under various operating conditions. These calculations help designers identify the most critical factors affecting reliability and make informed design choices to ensure **safe, long-lasting, and efficient systems**.

## 1. CHEMICAL CAPACITOR:

### Component Details

- Name: **CSR** (Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability)
- **Capacitor Style:** CSR – Capacitor, Fixed, Electrolytic (Solid Electrolyte, Tantalum); Specification: MIL-C-39003; Description: Established Reliability; Base Failure Rate ( $\lambda_p$ ): 0.00040 failures/ $10^6$  h;
- **Capacitance:** 10  $\mu$ F ;
- **Quality:** Established Reliability, Style D;
- **\*Capacitor Rated Voltage:** 25 V;
- **Applied (Operating) Voltage:** 15 V;
- **Effective Resistance Between Capacitor and Power Supply:** 100  $\Omega$

RESULTS AT DIFFERENT TEMPERATURES:  
TEMPERATURE : 25 °C

Environment: GB(Ground, Benign)

Results

| Parameter   | Value                                   | Comment                           |
|-------------|-----------------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000001 Failures/10 <sup>6</sup> Hours | Failure Rate                      |
| MTBF        | 7.360546e+11 hours                      | Mean Time Between Failures (MTBF) |
| FIT         | 0.001359 Failures/10 <sup>9</sup> Hours | Failures in Time                  |

Environment: GM(Ground, Mobile)

Results

| Parameter   | Value                                   | Comment                           |
|-------------|-----------------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000027 Failures/10 <sup>6</sup> Hours | Failure Rate                      |
| MTBF        | 3.680273e+10 hours                      | Mean Time Between Failures (MTBF) |
| FIT         | 0.027172 Failures/10 <sup>9</sup> Hours | Failures in Time                  |

Environment: NS (Naval, Sheltered)

Results

| Parameter   | Value                                   | Comment                           |
|-------------|-----------------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000010 Failures/10 <sup>6</sup> Hours | Failure Rate                      |
| MTBF        | 1.051507e+11 hours                      | Mean Time Between Failures (MTBF) |
| FIT         | 0.009510 Failures/10 <sup>9</sup> Hours | Failures in Time                  |

REMARKS:

- In the Ground Benign (GB) environment, the capacitor shows **excellent reliability** with the **lowest failure rate** and the **highest MTBF**, as the stable and mild conditions minimize thermal and mechanical stress.
- In the Ground Mobile (GM) environment, reliability slightly **decreases** due to **mechanical vibrations and varying temperature cycles**, which introduce minor stress on the electrolyte and internal connections.

- Under the **Naval Sheltered (NS)** condition, **humidity and salt exposure** cause further degradation, increasing the failure rate and reducing MTBF compared to GB.

## TEMPERATURE: 45 °C

**Environment: GB**(Ground, Benign)

### Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000002 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 5.097493e+11 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.001962 Failures/ $10^9$ Hours | Failures in Time                  |

**Environment: GM**(Ground, Mobile)

### Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000039 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 2.548746e+10 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.039235 Failures/ $10^9$ Hours | Failures in Time                  |

**Environment: NS** (Naval, Sheltered)

### Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000014 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 7.282132e+10 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.013732 Failures/ $10^9$ Hours | Failures in Time                  |

### REMARKS:

- At **Ground Benign (GB)**, the capacitor maintains **high reliability**, though the MTBF shows a slight reduction compared to 25 °C, indicating that temperature rise begins to influence electrolyte stability.
- In the **Ground Mobile (GM)** environment, the **failure rate increases notably** due to combined **thermal and mechanical stresses**, which accelerate chemical reactions inside the capacitor.
- Under the **Naval Sheltered (NS)** condition, exposure to **moisture and salt particles** further degrades performance, resulting in a higher failure rate and a clear drop in MTBF.

TEMPERATURE: 85 °C

Environment: GB(Ground, Benign)

Results

| Parameter   | Value                                   | Comment                           |
|-------------|-----------------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000004 Failures/10 <sup>6</sup> Hours | Failure Rate                      |
| MTBF        | 2.765231e+11 hours                      | Mean Time Between Failures (MTBF) |
| FIT         | 0.003616 Failures/10 <sup>9</sup> Hours | Failures in Time                  |

Environment: GM(Ground, Mobile)

Results

| Parameter   | Value                                   | Comment                           |
|-------------|-----------------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000072 Failures/10 <sup>6</sup> Hours | Failure Rate                      |
| MTBF        | 1.382615e+10 hours                      | Mean Time Between Failures (MTBF) |
| FIT         | 0.072327 Failures/10 <sup>9</sup> Hours | Failures in Time                  |

Environment: NS (Naval, Sheltered)

Results

| Parameter   | Value                                   | Comment                           |
|-------------|-----------------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000025 Failures/10 <sup>6</sup> Hours | Failure Rate                      |
| MTBF        | 3.950330e+10 hours                      | Mean Time Between Failures (MTBF) |
| FIT         | 0.025314 Failures/10 <sup>9</sup> Hours | Failures in Time                  |

REMARKS:

- In the **Ground Benign (GB)** environment, the capacitor still operates reliably but shows a **clear reduction in MTBF**, confirming that high temperature accelerates internal chemical degradation. Under **Ground Mobile (GM)** conditions, both **thermal and vibration stress** significantly increase the **failure rate**, leading to much lower reliability compared to GB.
- The **Naval Sheltered (NS)** environment exhibits **moderate reliability loss**, as moisture and thermal effects combine to raise the failure rate and shorten the operational lifetime.

**A) Environments at a fixed temperature (T = 85 °C) :**

| Environment           | $\lambda_p$ (Failures/ $10^6$ h) | MTBF (h)     | FIT (Failures/ $10^9$ h) |
|-----------------------|----------------------------------|--------------|--------------------------|
| GB (Ground, Benign)   | 0.000004                         | 2.765231e+11 | 0.003616                 |
| GM (Ground, Mobile)   | 0.000072                         | 1.382615e+10 | 0.072327                 |
| NS (Naval, Sheltered) | 0.000025                         | 3.950330e+10 | 0.025314                 |

**B) Temperatures in a fixed environment (Env = GB) :**

| Temperature (°C) | $\lambda_p$ (Failures/ $10^6$ h) | MTBF (h)     | FIT (Failures/ $10^9$ h) |
|------------------|----------------------------------|--------------|--------------------------|
| 25               | 0.000001                         | 7.360546e+11 | 0.001359                 |
| 45               | 0.000002                         | 5.097493e+11 | 0.001962                 |
| 85               | 0.000004                         | 2.765231e+11 | 0.003616                 |

**REMARKS:**

- The capacitor shows best reliability in the Ground Benign (GB) environment with the lowest failure rate (0.000004) and highest MTBF ( $2.76 \times 10^{11}$  h), confirming excellent stability under mild conditions. Ground Mobile (GM) and Naval Sheltered (NS) environments cause higher failure rates due to vibration, humidity, and corrosion, reducing overall reliability.
- As the temperature increases from 25 °C to 85 °C, the failure rate rises and MTBF drops, showing that thermal stress accelerates degradation. Overall, temperature and environment are dominant factors in capacitor reliability — with GB at lower temperatures providing maximum lifespan.

## VOLTAGE ANALYSIS – VOLTAGE IMPACT ON CHEMICAL CAPACITOR (CSR, TANTALUM, SOLID ELECTROLYTE):

**Parameters:** Capacitor Style: CSR – Capacitor, Fixed, Electrolytic (Solid Electrolyte, Tantalum); Capacitance = 10  $\mu\text{F}$ ; Rated Voltage = 25 V; Quality = Style D (Established Reliability); Effective Resistance Between Capacitor and Power Supply: 100  $\Omega$ .

**Environment** = GB (Ground Benign);

**Temperature:** 85  $^{\circ}\text{C}$ ;

| Applied Voltage (V) | Failure Rate ( $\lambda_p$ ) (Failures/ $10^6$ h) | MTBF (hours)              | FIT (Failures/ $10^9$ h) | Observation                                              |
|---------------------|---------------------------------------------------|---------------------------|--------------------------|----------------------------------------------------------|
| 8V                  | 0.000001                                          | $8.379296 \times 10^{11}$ | 0.001193                 | Lowest stress, highest MTBF (best reliability)           |
| 12V                 | 0.000001                                          | $8.194953 \times 10^{11}$ | 0.001220                 | Slight reduction in MTBF, small increase in FIT          |
| 16V                 | 0.000007                                          | $1.384127 \times 10^{11}$ | 0.007225                 | Higher stress, increased failure rate, lower reliability |

### REMARKS:

- At **8V**, the capacitor experiences **minimal electrical stress**, leading to the **highest MTBF** and best overall reliability since the applied voltage is well below the rated value.
- At **12V**, reliability **slightly decreases** as the voltage approaches the rated limit, causing a small increase in electrical stress and internal heating effects.
- At **16V**, the **failure rate increases sharply**, confirming that operating beyond 30% of the rated voltage significantly accelerates degradation, reducing the capacitor's lifetime.

## CERAMIC CAPACITOR:

### Component Details

- **Name:** CKR (Capacitor, Fixed, Ceramic Dielectric, General Purpose, Established Reliability); **Capacitor Style:** CKR – Capacitor, Fixed, Ceramic Dielectric; **Specification:** MIL-C-123; **Description:** Established Reliability; **Base Failure Rate ( $\lambda_p$ ):** 0.000003 failures/ $10^6$  h;
- **Capacitance:** 0.1  $\mu$ F;
- **Quality:** Established Reliability, Style D
- **Capacitor Rated Voltage:** 50 V
- **Applied (Operating) Voltage:** 40 V
- **Temperature:** 25 °C, 45 °C, and 85 °C
- **Effective Resistance Between Capacitor and Power Supply:** 100  $\Omega$  (only considered for power loss estimation)
- **Environment:** GB (Ground Benign), GM (Ground Mobile), and NS (Naval Sheltered)

### RESULTS AT DIFFERENT TEMPERATURES:

#### TEMPERATURE : 25 °C

**Environment:** GB(Ground, Benign)

#### Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000003 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 3.687119e+11 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.002712 Failures/ $10^9$ Hours | Failures in Time                  |

**Environment:** GM(Ground, Mobile)

#### Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000054 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 1.843560e+10 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.054243 Failures/ $10^9$ Hours | Failures in Time                  |

Environment: NS (Naval, Sheltered)

Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000019 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 5.267313e+10 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.018985 Failures/ $10^9$ Hours | Failures in Time                  |

REMARKS:

- At **Ground Benign (GB)**, the capacitor achieves **very high reliability** with the lowest failure rate and highest MTBF, showing that it performs stably under mild and controlled ambient conditions.
- Under the **Ground Mobile (GM)** environment, the **failure rate increases sharply**, mainly due to **mechanical vibrations and temperature cycling**, which can cause micro-cracks or stress in the ceramic dielectric.
- In the **Naval Sheltered (NS)** condition, the reliability slightly drops compared to GB, likely because of **humidity and salt-trace effects** that influence dielectric insulation and accelerate minor leakage paths.

TEMPERATURE: 45 °C

Environment: GB(Ground, Benign)

Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000006 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 1.564573e+11 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.006392 Failures/ $10^9$ Hours | Failures in Time                  |

Environment: GM(Ground, Mobile)

Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000128 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 7.822866e+9 hours               | Mean Time Between Failures (MTBF) |
| FIT         | 0.127830 Failures/ $10^9$ Hours | Failures in Time                  |

Environment: NS (Naval, Sheltered)

## Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000045 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 2.235105e+10 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.044741 Failures/ $10^9$ Hours | Failures in Time                  |

### REMARKS:

- At **Ground Benign (GB)**, the capacitor maintains **strong reliability** even at moderate temperature, meaning its dielectric material performs well under low electrical and thermal stress. In the **Ground Mobile (GM)** environment, the higher **failure rate** suggests that **vibrations and motion** create additional mechanical strain, reducing the effective lifetime of the capacitor.
- Under the **Naval Sheltered (NS)** environment, the reliability decreases moderately, which I interpret as the **combined impact of humidity and salt traces**, slightly accelerating the aging process.

**TEMPERATURE:** 85 °C

**Environment:** GB(Ground, Benign)

## Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000027 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 3.754946e+10 hours              | Mean Time Between Failures (MTBF) |
| FIT         | 0.026632 Failures/ $10^9$ Hours | Failures in Time                  |

**Environment:** GM(Ground, Mobile)

## Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000533 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 1.877473e+9 hours               | Mean Time Between Failures (MTBF) |
| FIT         | 0.532631 Failures/ $10^9$ Hours | Failures in Time                  |

**Environment:** NS (Naval, Sheltered)

## Results

| Parameter   | Value                           | Comment                           |
|-------------|---------------------------------|-----------------------------------|
| $\lambda_p$ | 0.000186 Failures/ $10^6$ Hours | Failure Rate                      |
| MTBF        | 5.364208e+9 hours               | Mean Time Between Failures (MTBF) |
| FIT         | 0.186421 Failures/ $10^9$ Hours | Failures in Time                  |

### REMARKS:

- In the Ground Benign (GB) environment, the capacitor exhibits moderate reliability, though the higher temperature causes a noticeable drop in MTBF compared to lower temperatures, showing thermal sensitivity of ceramic materials. Under the Ground Mobile (GM) condition, the failure rate increases sharply due to the combined effects of temperature and vibration, which can induce mechanical stress and micro-cracks in the dielectric layer.
- In the Naval Sheltered (NS) environment, humidity and salt exposure at elevated temperature lead to further degradation, resulting in the highest failure rate and lowest MTBF, marking the worst performance among all environments.

### A) Environments at a fixed temperature (T = 45 °C):

| Environment                  | $\lambda_p$ (Failures/ $10^6$ h) | MTBF (h)     | FIT (Failures/ $10^9$ h) |
|------------------------------|----------------------------------|--------------|--------------------------|
| <b>GB</b> (Ground, Benign)   | 0.000006                         | 1.564573e+11 | 0.006392                 |
| <b>GM</b> (Ground, Mobile)   | 0.000128                         | 7.822866e+9  | 0.127830                 |
| <b>NS</b> (Naval, Sheltered) | 0.000045                         | 2.235105e+10 | 0.044741                 |

### B) Temperatures at a fixed environment (Env = NS):

| Temperature (°C) | $\lambda_p$ (Failures/ $10^6$ h) | MTBF (h)     | FIT (Failures/ $10^9$ h) |
|------------------|----------------------------------|--------------|--------------------------|
| 25               | 0.000019                         | 5.267313e+10 | 0.018985                 |
| 45               | 0.000045                         | 2.235105e+10 | 0.044741                 |
| 85               | 0.000186                         | 5.364208e+9  | 0.186421                 |

## REMARKS:

- At a fixed temperature (45 °C), the Ground Benign (GB) environment provides the highest reliability with the lowest failure rate, while Ground Mobile (GM) shows a strong decrease in MTBF due to vibration and motion stress. The Naval Sheltered (NS) environment performs better than GM but worse than GB, suggesting moderate impact from humidity and mild corrosion.
- When temperature increases within a fixed environment (NS), the failure rate rises exponentially from 25 °C to 85 °C, confirming the temperature dependence of dielectric degradation. The MTBF consistently decreases as temperature rises, highlighting that ceramic capacitors are thermally sensitive and more prone to stress-induced failure at elevated temperatures.

## TRANSISTOR RELIABILITY ANALYSIS:

### Objective:

The aim of this experiment is to evaluate the reliability of a MOSFET transistor under different environmental and temperature conditions using the **MIL-HDBK-217F** standard. The goal is to determine the **failure rate ( $\lambda_b$ )**, **FIT (Failures in Time)**, and **MTBF (Mean Time Between Failures)** for three environments:

- Ground Benign (GB)
- Ground Mobile (GM)
- Naval Sheltered (NS)

and for three junction temperatures: **40 °C, 85 °C, and 125 °C**.

### Methodology:

The reliability prediction for the MOSFET transistor is performed using the following equation:

$$\lambda_b = \pi_E \times \pi_Q \times \pi_T \times \lambda_G$$

where:

- $\lambda_G$ : Generic base failure rate (MOSFET = 0.004 failures/10<sup>6</sup> h)
- $\pi_E$ : Environment factor (1 = GB, 5 = GM, 9 = NS)
- $\pi_Q$  : Quality factor (1 = Commercial grade)
- $\pi_T$  : Temperature factor (depends on junction temperature, calculated using exponential model)

The **Mean Time Between Failures (MTBF)** is calculated as:

And  $FIT = \lambda_b \times 1000$

since FIT expresses failures per  $10^9$  hours.

### Parameters:

| Parameter             | Symbol      | Value / Description         |
|-----------------------|-------------|-----------------------------|
| Generic Failure Rate  | $\lambda_G$ | 0.004 failures / $10^6$ h   |
| Activation Energy     | $E_a$       | 0.3 eV                      |
| Quality Factor        | $\pi_Q$     | 1 (Commercial)              |
| Reference Temperature | $T_{ref}$   | 25 °C (298 K)               |
| Boltzmann's Constant  | k           | $8.617 \times 10^{-5}$ eV/K |

### Results and Discussion:

#### A) Temperature Factor Calculation ( $\pi_T$ )

The temperature factor  $\pi_T$  is calculated using:

$$\pi_T = \exp \left[ -\frac{E_a}{k} \left( \frac{1}{T_j} - \frac{1}{T_{ref}} \right) \right]$$

#### Resulting $\pi_T$ values:

| Temperature (°C) | Temperature (K) | $\pi_T$ |
|------------------|-----------------|---------|
| 40               | 313.15          | 1.7495  |
| 85               | 358.15          | 7.0725  |
| 125              | 398.15          | 18.7803 |

### Ground Benign:

| MOSFET Reliability Calculator (MIL-HDBK-217F)                                    |  |  |  | $T_j$ (°C)                                                   | $T_j$ (K) | $\pi_T$  | $\lambda_b$ (failures/million hrs) | FIT  |
|----------------------------------------------------------------------------------|--|--|--|--------------------------------------------------------------|-----------|----------|------------------------------------|------|
| Input Parameters                                                                 |  |  |  | 40                                                           | 313,15    | 1,749502 | 0,006998009                        | 7,0  |
| Generic Failure Rate ( $\lambda_G$ )                                             |  |  |  | 85                                                           | 358,15    | 7,072509 | 0,028290035                        | 28,3 |
| Environment Factor ( $\pi_E$ )                                                   |  |  |  | 125                                                          | 398,15    | 18,78029 | 0,075121164                        | 75,1 |
| Quality Factor ( $\pi_Q$ )                                                       |  |  |  |                                                              |           |          |                                    |      |
| Junction Temperature ( $T_j$ ) in °C                                             |  |  |  |                                                              |           |          |                                    |      |
| Activation Energy ( $E_a$ ) in eV                                                |  |  |  |                                                              |           |          |                                    |      |
| $\pi_T$                                                                          |  |  |  |                                                              |           |          |                                    |      |
| Example                                                                          |  |  |  |                                                              |           |          |                                    |      |
| 0,004                                                                            |  |  |  |                                                              |           |          |                                    |      |
| 1 (Ground Benign)                                                                |  |  |  |                                                              |           |          |                                    |      |
| 1 (Commercial)                                                                   |  |  |  |                                                              |           |          |                                    |      |
| 80 °C                                                                            |  |  |  |                                                              |           |          |                                    |      |
| 0,3 eV                                                                           |  |  |  |                                                              |           |          |                                    |      |
| 6,16                                                                             |  |  |  |                                                              |           |          |                                    |      |
| For ( $\pi_E$ ):                                                                 |  |  |  | The formula is: $\pi_T = \exp[-E_a/k * (1/T_j - 1/T_{ref})]$ |           |          |                                    |      |
| Data Validation → List → Source: 1.0, 2.0, 4.0, 5.0, 9.0, 13.0, 15.0             |  |  |  | Where:                                                       |           |          |                                    |      |
| Label each value with the corresponding environment (e.g., 1.0 = Ground, Benign) |  |  |  | $E_a$ = Activation energy (eV), typically 0.3 eV for MOSFETs |           |          |                                    |      |
| Select Ground Benign : 1                                                         |  |  |  | $k$ = Boltzmann's constant ( $8.617 \times 10^{-5}$ eV/K)    |           |          |                                    |      |
|                                                                                  |  |  |  | $T_j$ = Junction temperature in Kelvin (K)                   |           |          |                                    |      |
|                                                                                  |  |  |  | $T_{ref}$ = Reference temperature, usually 298 K (25°C)      |           |          |                                    |      |
|                                                                                  |  |  |  |                                                              |           |          |                                    |      |
|                                                                                  |  |  |  |                                                              |           |          |                                    |      |
|                                                                                  |  |  |  |                                                              |           |          |                                    |      |
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|                                                                                  |  |  |  |                                                              |           |          |                                    |      |
|                                                                                  |  |  |  |                                                              |           |          |                                    |      |
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## Ground Mobile:

| MOSFET Reliability Calculator (MIL-HDBK-217F)                                    |  |  | Tj (°C)                                                   | Tj (K) | πT       | λb (failures/million hrs) | FIT       |
|----------------------------------------------------------------------------------|--|--|-----------------------------------------------------------|--------|----------|---------------------------|-----------|
| Input Parameters                                                                 |  |  | 40                                                        | 313,15 | 1,749502 | 0,034990047               | 35,0      |
| Generic Failure Rate (λG)                                                        |  |  | 85                                                        | 358,15 | 7,072509 | 0,141450175               | 141,5     |
| Environment Factor (πE)                                                          |  |  | 125                                                       | 398,15 | 18,78029 | 0,375605821               | 375,6     |
| Quality Factor (πQ)                                                              |  |  |                                                           |        |          |                           |           |
| Junction Temperature (Tj) in °C                                                  |  |  |                                                           |        |          |                           |           |
| Activation Energy (Ea) in eV                                                     |  |  |                                                           |        |          |                           |           |
| πT                                                                               |  |  |                                                           |        |          |                           |           |
|                                                                                  |  |  |                                                           |        |          |                           |           |
| For (πE):                                                                        |  |  | The formula is: $\pi T = \exp[-Ea/k * (1/Tj - 1/Tref)]$   |        |          |                           |           |
| Data Validation → List → Source: 1.0, 2.0, 4.0, 5.0, 9.0, 13.0, 15.0             |  |  | Where:                                                    |        |          |                           |           |
| Label each value with the corresponding environment (e.g., 1.0 = Ground, Benign) |  |  | Ea = Activation energy (eV), typically 0.3 eV for MOSFETs |        |          |                           |           |
| Select Ground Mobile: 5                                                          |  |  | k = Boltzmann's constant (8.617 × 10 <sup>-5</sup> eV/K)  |        |          |                           | 8,62E-05  |
| For (πQ):                                                                        |  |  | Tj = Junction temperature in Kelvin (K)                   |        |          |                           | 80 353,15 |
| Data Validation → List → Source: 0.3, 0.7, 1.0, 2.0                              |  |  | Tref = Reference temperature, usually 298 K (25°C)        |        |          |                           | 25 298,15 |
| Label each value with the quality level                                          |  |  |                                                           |        |          |                           |           |
| Select: 1                                                                        |  |  |                                                           |        |          |                           |           |
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### Remarks:

- The temperature factor ( $\pi T$ ) increases exponentially with temperature — from 1.75 at 40 °C to 18.78 at 125 °C, indicating that transistor reliability is highly temperature-dependent due to accelerated semiconductor degradation. In all environments, failure rate ( $\lambda_b$ ) and FIT values rise sharply with temperature, while MTBF decreases drastically, confirming that thermal stress is the dominant cause of failure.
- The Ground Benign (GB) environment shows the highest reliability, followed by Naval Sheltered (NS), whereas Ground Mobile (GM) exhibits the worst performance due to mechanical vibration and dynamic thermal cycling. Overall, the transistor's lifetime reduces exponentially beyond 85 °C, emphasizing the need for effective cooling or thermal management in high-temperature electronic systems.

### Temperature Effect (Environment Fixed – Ground Benign):

When evaluating the effect of temperature under a constant **Ground Benign (GB)** environment, the transistor shows a clear degradation trend with increasing junction temperature. The failure rate ( $\lambda_b$ ) rises sharply from **0.00699 fail/10<sup>6</sup> h at 40 °C** to **0.02829 fail/10<sup>6</sup> h at 85 °C**, and further to **0.07512 fail/10<sup>6</sup> h at 125 °C**. Consequently, the **FIT** increases from **7.0 to 75.1 failures/10<sup>9</sup> h**, while the **MTBF** drastically decreases from **1.43 × 10<sup>8</sup> h to 1.33 × 10<sup>7</sup> h**. This demonstrates that **temperature is a dominant factor** in reducing transistor reliability due to accelerated semiconductor degradation and increased leakage currents at high thermal stress.

### Environment Effect (Temperature Fixed – 85 °C):

At a fixed junction temperature of **85 °C**, the reliability varies noticeably across different environmental conditions. Under **Ground Benign (GB)**, the transistor has the lowest failure rate ( $\lambda_b = 0.02829 \text{ fail/10}^6 \text{ h}$ ) and the highest MTBF (**3.54 × 10<sup>7</sup> h**). As the environment becomes more severe—**Ground Mobile (GM)** and **Naval Sheltered (NS)**—the failure rate increases to **0.14145** and **0.25461 fail/10<sup>6</sup> h**, respectively, while MTBF drops to **7.07 × 10<sup>6</sup> h** and **3.93 × 10<sup>6</sup> h**. This indicates that environmental stresses such as vibration, humidity, and mechanical shocks significantly affect reliability, making **Ground Benign** the most stable operating environment.

### SUMMARY:

From the analysis:

- The transistor (MOSFET) demonstrates **maximum reliability** at 40 °C in a **Ground Benign** environment with an MTBF of **1.43 × 10<sup>8</sup> hours**.
- Reliability significantly degrades at higher temperatures and harsher environments.
- The study confirms that **temperature and environment are the dominant factors** in transistor reliability according to MIL-HDBK-217F.

## CONCLUSION:

- The overall reliability analysis of electronic components — chemical capacitors, ceramic capacitors, and transistors — demonstrates that **temperature, electrical stress, and environmental conditions are the dominant factors** influencing component lifetime. Across all simulations, **Ground Benign (GB)** environments consistently provided the **highest reliability and MTBF**, owing to minimal vibration, humidity, and thermal variation. In contrast, **Ground Mobile (GM)** and **Naval Sheltered (NS)** conditions introduced mechanical, moisture, and corrosion stresses, which significantly increased the **failure rate ( $\lambda_p/\lambda_b$ )** and reduced **MTBF**. Similarly, increasing temperature or operating voltage accelerated internal degradation, confirming the strong thermal sensitivity of electrolytic and semiconductor devices. Overall, the study highlights that maintaining **lower operating temperatures, moderate voltages, and controlled environments** is essential for achieving **stable, long-lasting, and efficient system performance** in electronic design.